

Which areas of your business do you see as most critical for inaccurate AI analyses to lead to detrimental or very costly effects?

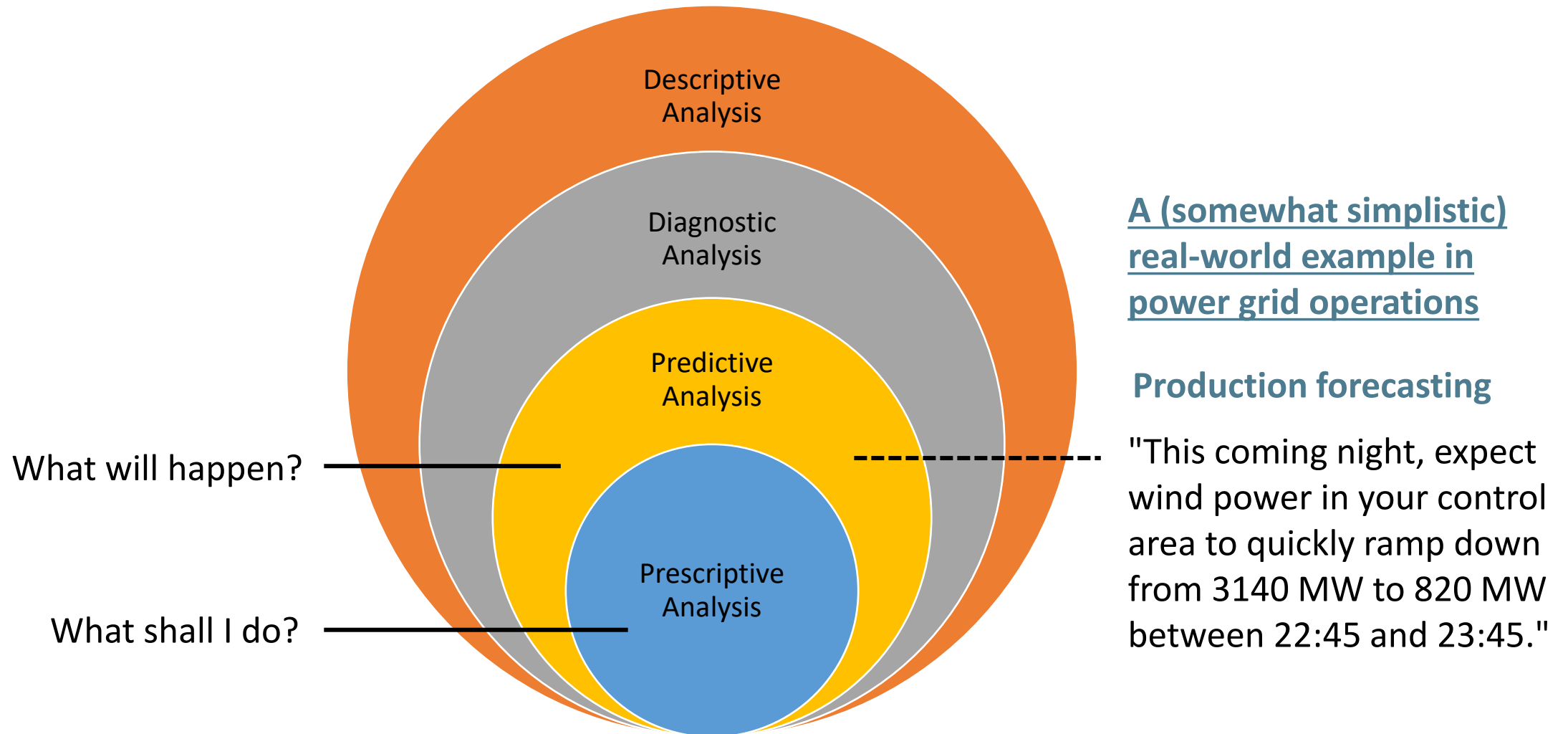


Interactive Session

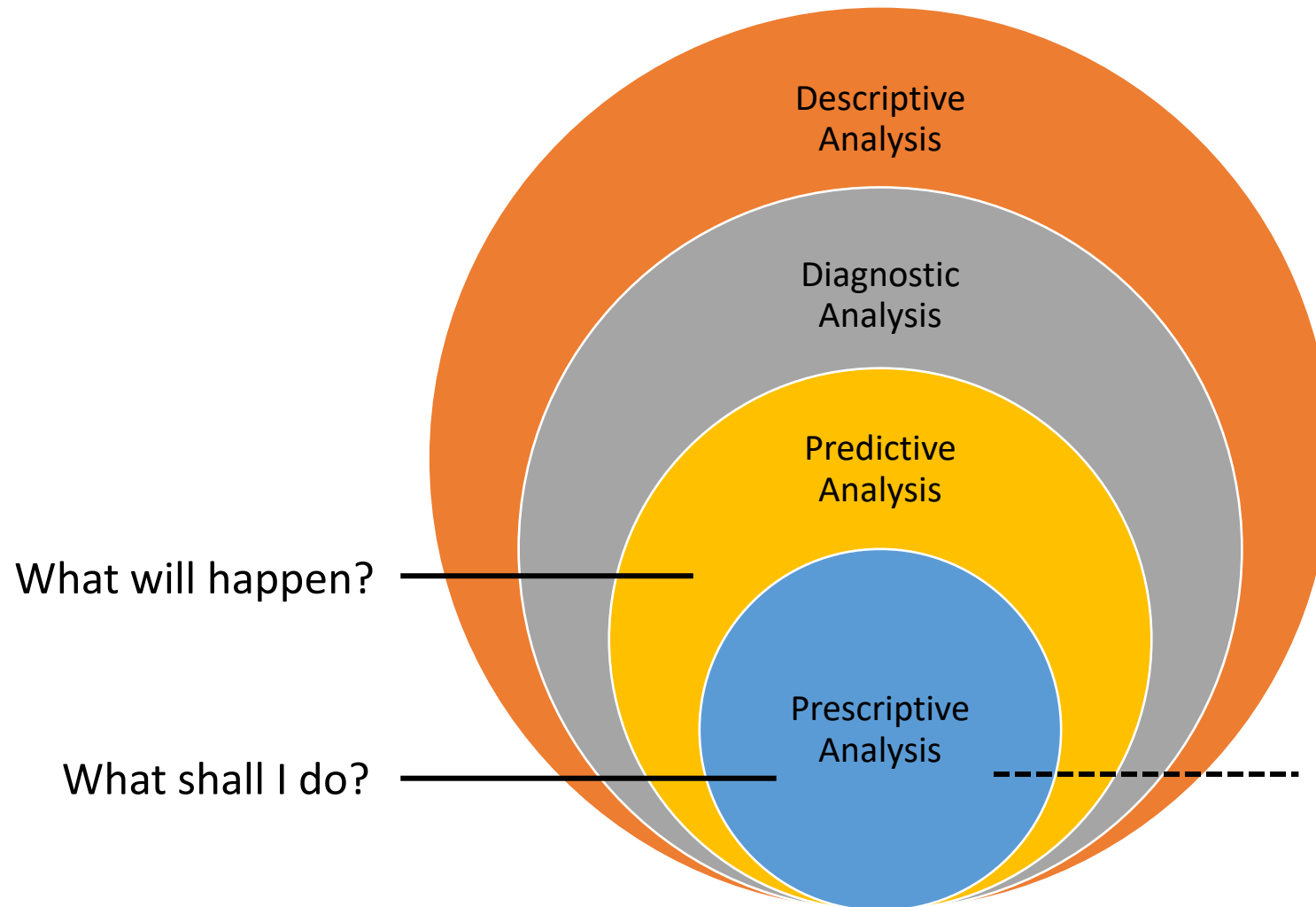
Data Science and AI for the energy system

- Implications of AI in forecasting and decision making
- Expert survey
- Session discussion

Implications of AI in forecasting and decision making



Implications of AI in forecasting and decision making

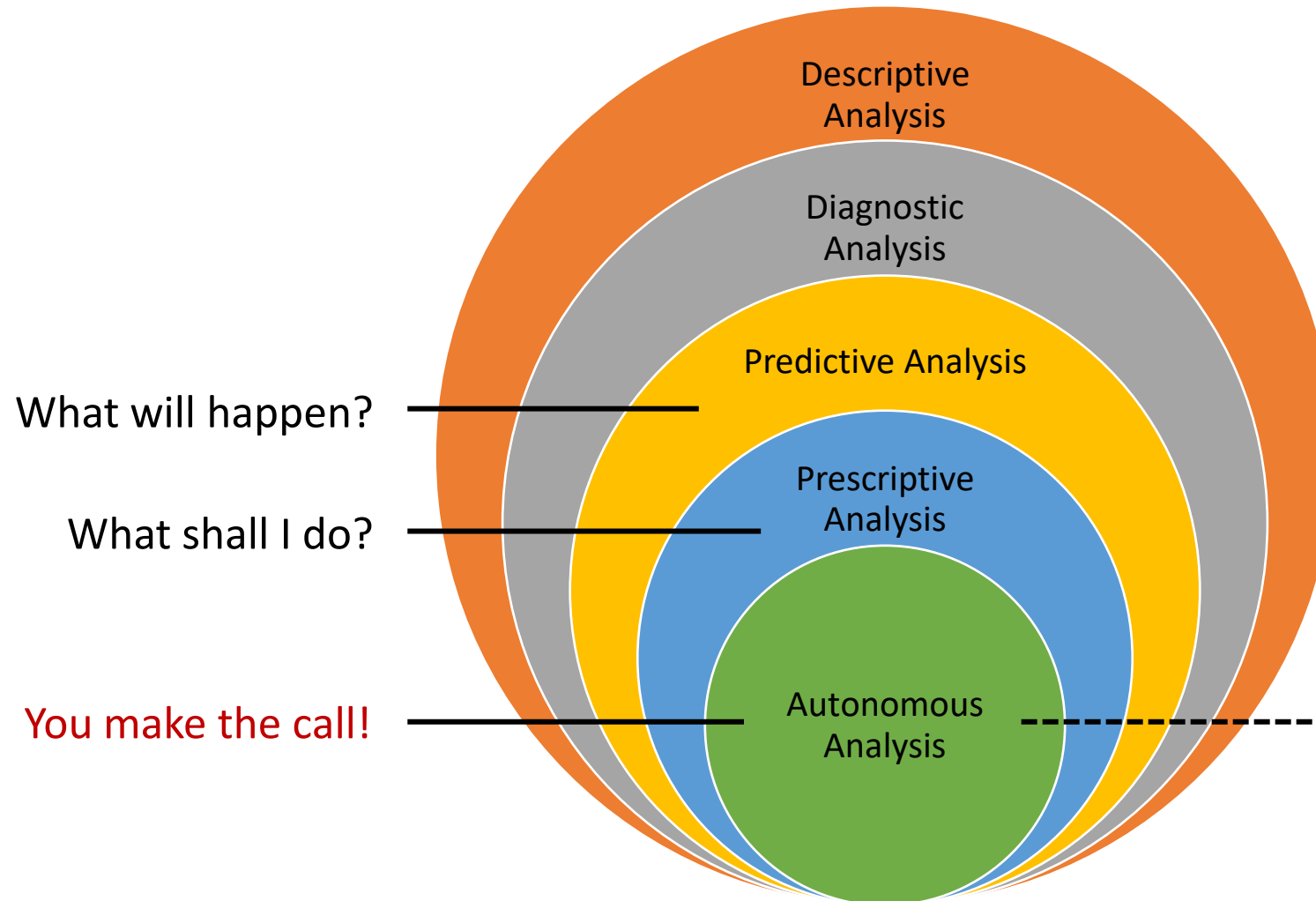


A (somewhat simplistic) real-world example in power grid operations

Decision Support

"This coming night, starting at 22:45, request reserves on the order of 2.3 GW to compensate a shift in wind power production."

Implications of AI in forecasting and decision making

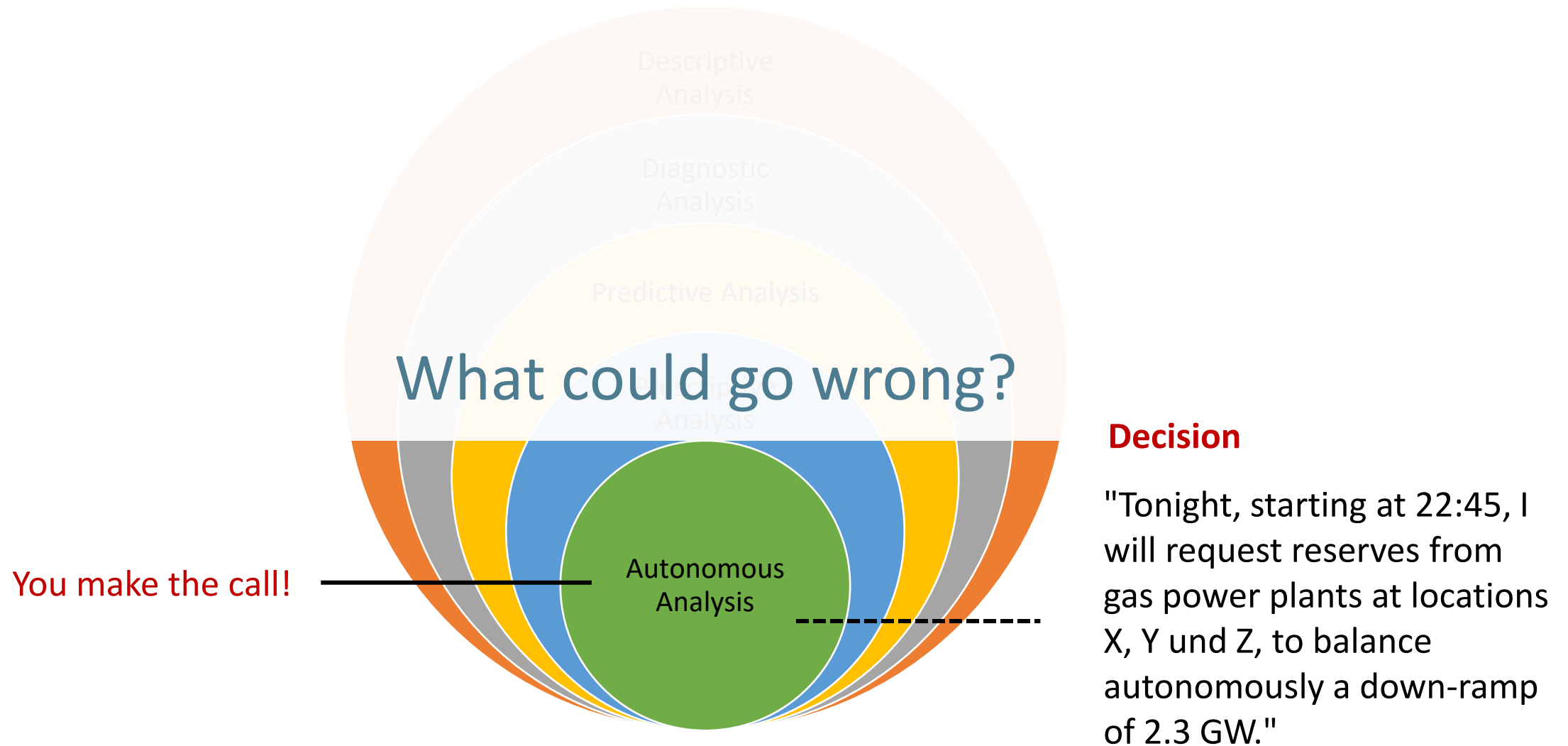


A (somewhat simplistic) real-world example in power grid operations

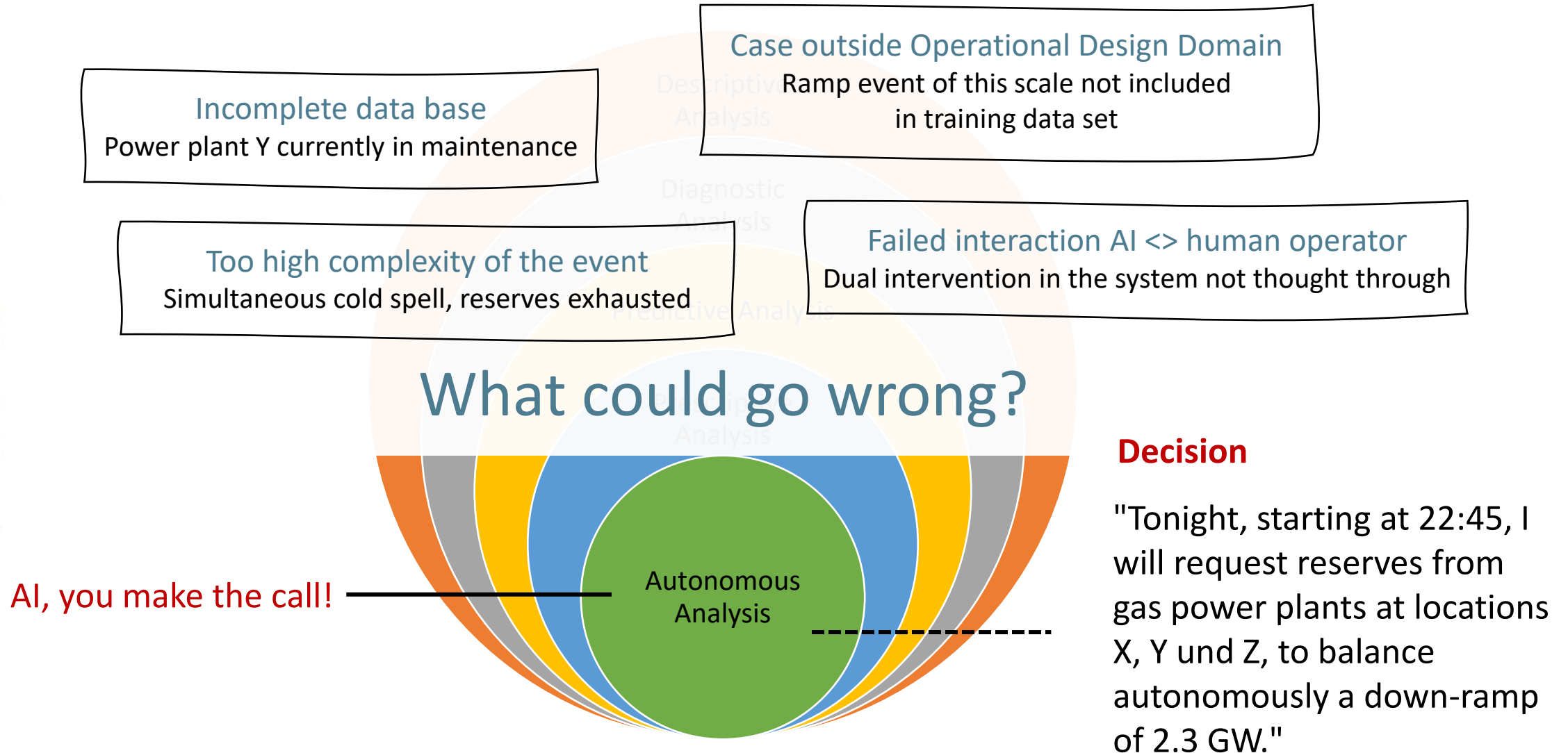
Decision

"Tonight, starting at 22:45, I will request reserves from gas power plants at locations X, Y und Z, to balance autonomously a down-ramp of 2.3 GW."

Implications of AI in forecasting and decision making



Implications of AI in forecasting and decision making

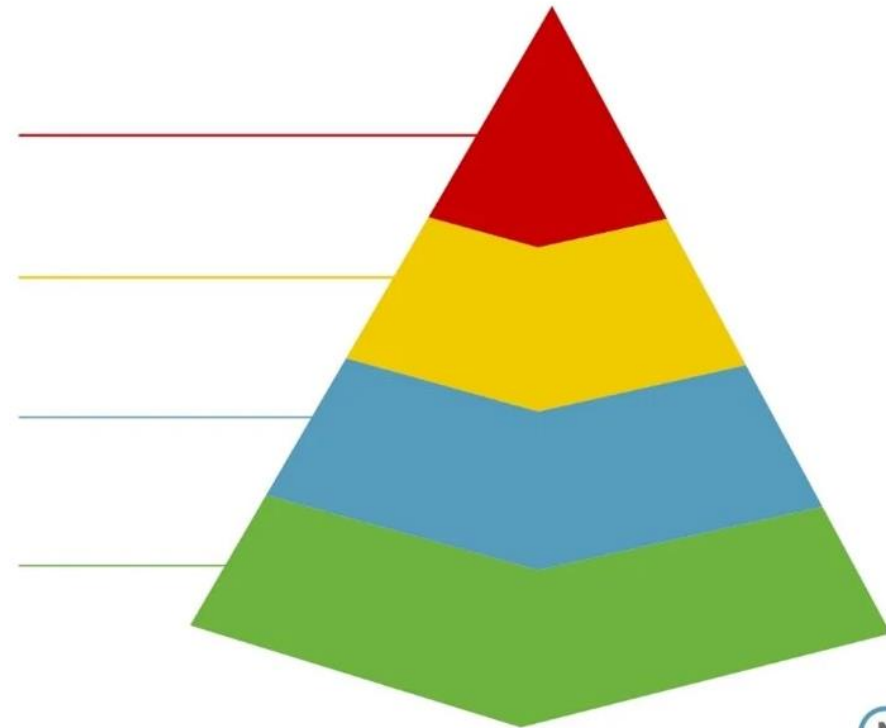
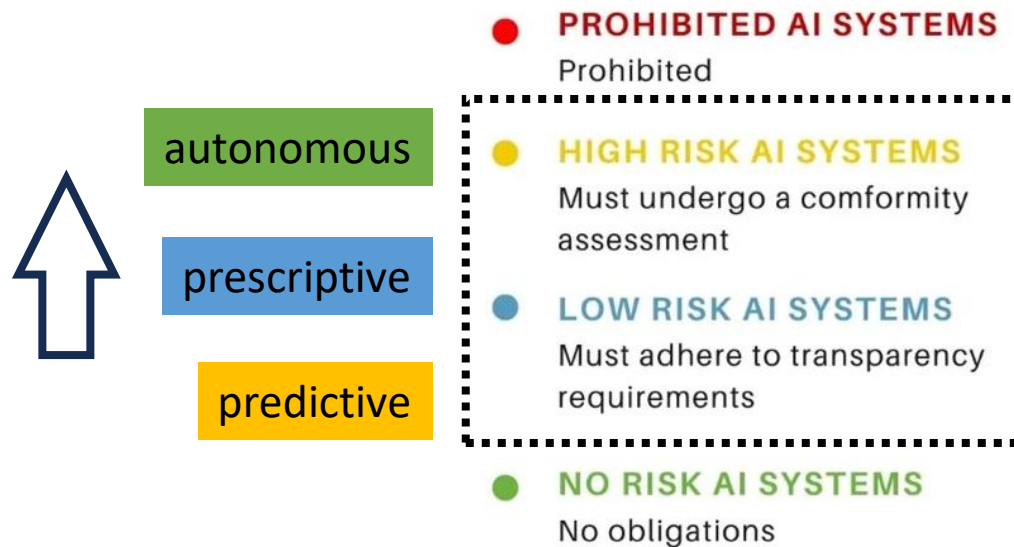


AI risks in forecasting and decision making



EU Artificial Intelligence Act

<https://artificialintelligenceact.eu>

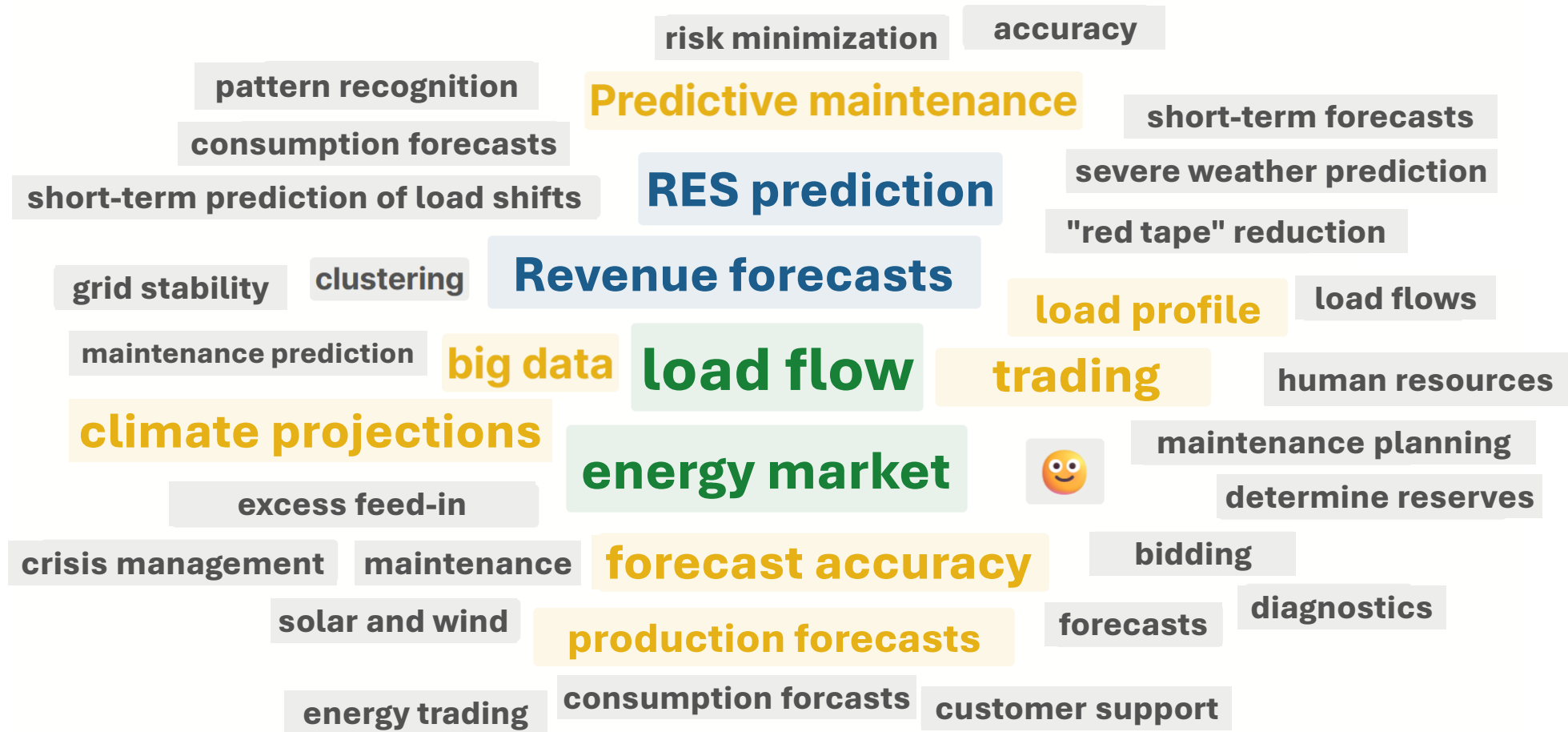


Expert survey: AI potentials

For which areas in the energy sector do you see the greatest potential for AI to lead to faster and more accurate analyses?

Give individual answers. Multiple answers are possible.

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Expert survey: AI risks

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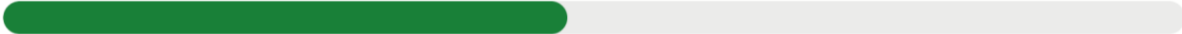
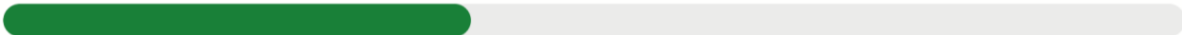
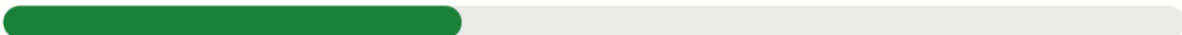
Expert survey: AI outlook

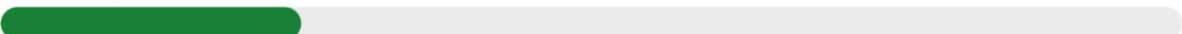
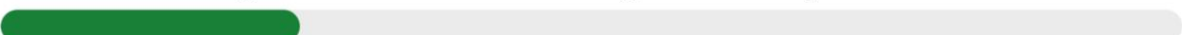
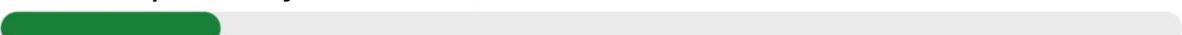
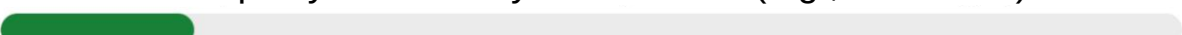
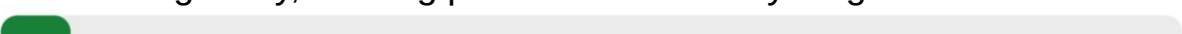
What measures are needed to equip your company / your research institution for a deeper integration of AI?

Choose from multiple answers and sort them by importance.

What measures are needed to prepare your company / your research institution for a deeper integration of AI?

1. AI experts within the organization

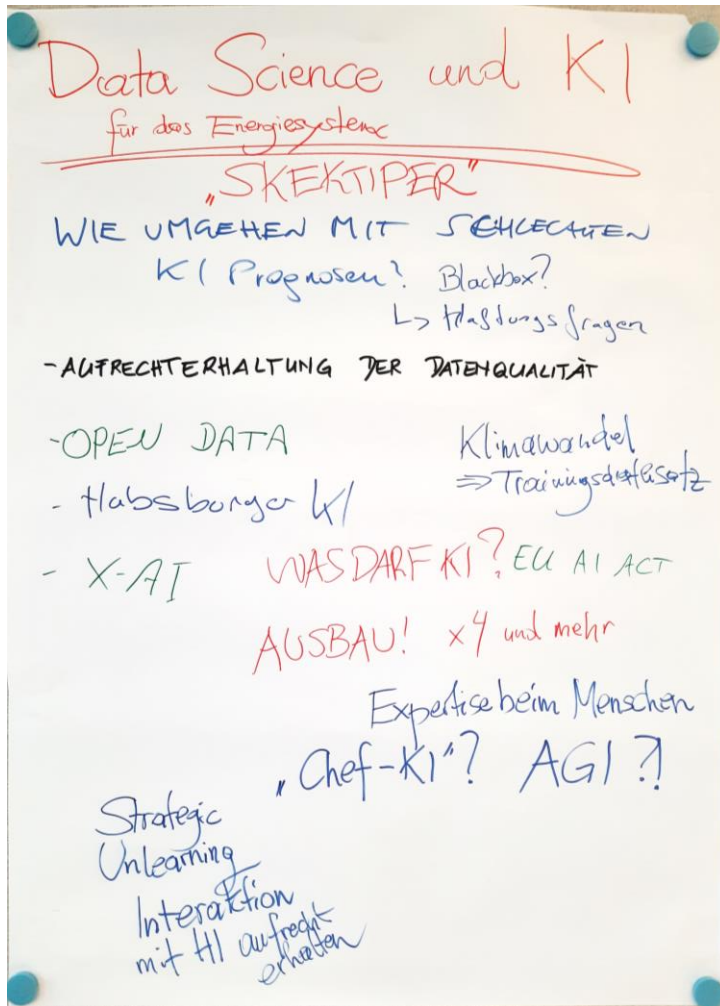
2. Training courses & education

3. Exchange with AI experts from other industry sectors / research

4. Research projects to explore AI potentials

5. In-depth technical workshops

6. Ready-for-use solutions by established (software and forecast) providers

7. Certification of existing AI-driven systems

8. In-depth study of the EU AI Act

9. AI roadmaps by the industry associations (e.g., ENTSO-E)

10. Nothing really, existing providers offer everything we need


Session discussion



Session discussion



Data Science and AI for the energy system

"AI skepticism"

How to cope with bad

AI forecasts? Black box?

-> Liability!?

- Maintenance of data quality

- Open Data

Climate change

- Habsburgian AI

=> training data set

- X-AI

What do we permit AI to do?

EU AI Act

Expansion of renewables! x 4 or more

Expertise must remain with humans

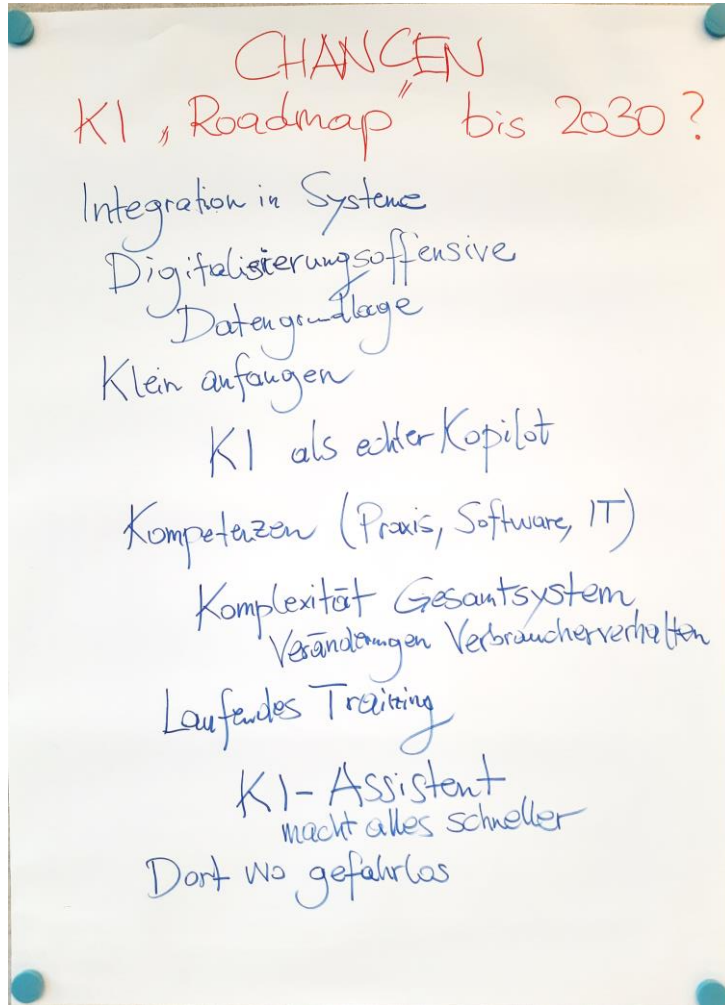
"AI boss"? AGI?!

Strategic

Unlearning

Maintain interaction
with human (HI)

Session discussion



OPPORTUNITIES

AI "Roadmap" towards 2030?

Integration into systems

Digitalization campaign

Data foundations

Start small

AI as an actual co-pilot

Competences (operations, software, IT)

Overall complexity of the system

Changes in consumer behaviour

Continuous training

AI assistant

makes everything faster

AI where it's safe

Interactive Session

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